

Image quality safety model for the safety of the intended functionality in highly automated agricultural machines

Changjoo Lee^{a,b,*}, Simon Schätzle^b, Stefan Andreas Lang^b, Timo Oksanen^a

^a Technical University of Munich, Chair of Agrimechatronics & Munich Institute of Robotics and Machine Intelligence (MIRMI), 85354 Freising, Germany

^b Sensor-Technik Wiedemann GmbH, Bärenwald 6, 87600 Kaufbeuren, Germany

ARTICLE INFO

Keywords:

Perception system
Safety of the intended functionality
Functional insufficiency
Highly automated agricultural machine
Image quality safety model
ISO 18497

ABSTRACT

Achieving safe and reliable environmental perception is crucial for the success of highly automated or even autonomous agricultural machinery. However, developing such a system is challenging due to the inherent limitations of perception sensors. In certain conditions, these sensors may fail to capture accurate data, leading to erroneous perceptions of the environment and potentially compromising safety. Monitoring the functional insufficiencies of the measurement data is crucial for ensuring the safety and reliability of perception systems.

This article introduces ISO standards, which provide guidelines for ensuring functional safety in highly automated mobile machines and vehicles. It also proposes an Image Quality Safety Model (IQSM) for monitoring the safety of the intended functionality in perception systems. The IQSM estimates the confidence level with which a camera can safely perform a specific object detection task. If the confidence level falls below a pre-defined threshold, the IQSM can trigger actions, alert operators, and prevent potential safety hazards. The IQSM exhibits remarkable performance, achieving a validation accuracy of about 90%, demonstrating its ability to effectively distinguish the safety of the intended functionality under a variety of image quality conditions.

1. Introduction

In line with the advent of autonomous vehicles, agricultural machines have used high-tech systems and sensors and have been able to perform certain tasks automatically for decades (Edan et al., 2009; Zhang, 2016). This integration has enabled the automated execution of certain tasks with minimal human intervention, resulting in increased efficiency and productivity (Christiansen, 2017; Linenko et al., 2019; Portugal et al., 2015). However, the driver still needs to monitor the operation for safety reasons. To perform highly automated operations without human intervention, the machines must gather and process environmental information to avoid hazardous situations. Thus, it is necessary to introduce a perception system that provides the information required for the safe control of machines, which poses great potential for reducing hazards (ISO, 2018a).

According to ISO 18497, a highly automated agricultural machine (HAAM) must be equipped with a perception system capable of detecting, under foreseeable operating conditions, (1) objects in its path, (2) persons approaching it, and (3) its position relative to detected obstacles (ISO, 2018a). Artificial intelligence (AI) and machine learning (ML)-powered perception systems are being used to develop perception

systems for HAAMs (ISO, 2024).

However, realising a highly functionally safe and reliable perception system presents significant challenges (Doran et al., 2020; Goelles et al., 2020; Marti et al., 2019). There is no perception sensor that works perfectly in all situations. Performance insufficiencies in perception sensors or systems can contribute to hazardous behaviour and may result in the inability to prevent, detect, and mitigate reasonably foreseeable misuse (ISO, 2022). The use of AI and ML in perception systems is accompanied by uncertainties in AI/ML model outcomes due to incomplete representations of the input domain during training, including factors such as seasonal variations and unforeseen inputs, as well as the inherent limitations of AI/ML models. Dealing with such uncertainties is crucial, especially when the actions of these systems have the potential to harm humans or the environment (ISO, 2024; Kläs and Vollmer, 2018).

In a variety of environmental conditions, data collected by perception sensors may be affected by various kinds of quality issues due to thick fog, heavy rain, etc. The actual level of uncertainty in the outcome of an AI/ML model is thus affected by the quality of the input runtime data on which it is currently applied (ISO, 2024). For instance, as shown in Fig. 1, a camera may face performance limitations in low-light

* Corresponding author.

E-mail address: changjoo.lee@wiedemann-group.com (C. Lee).

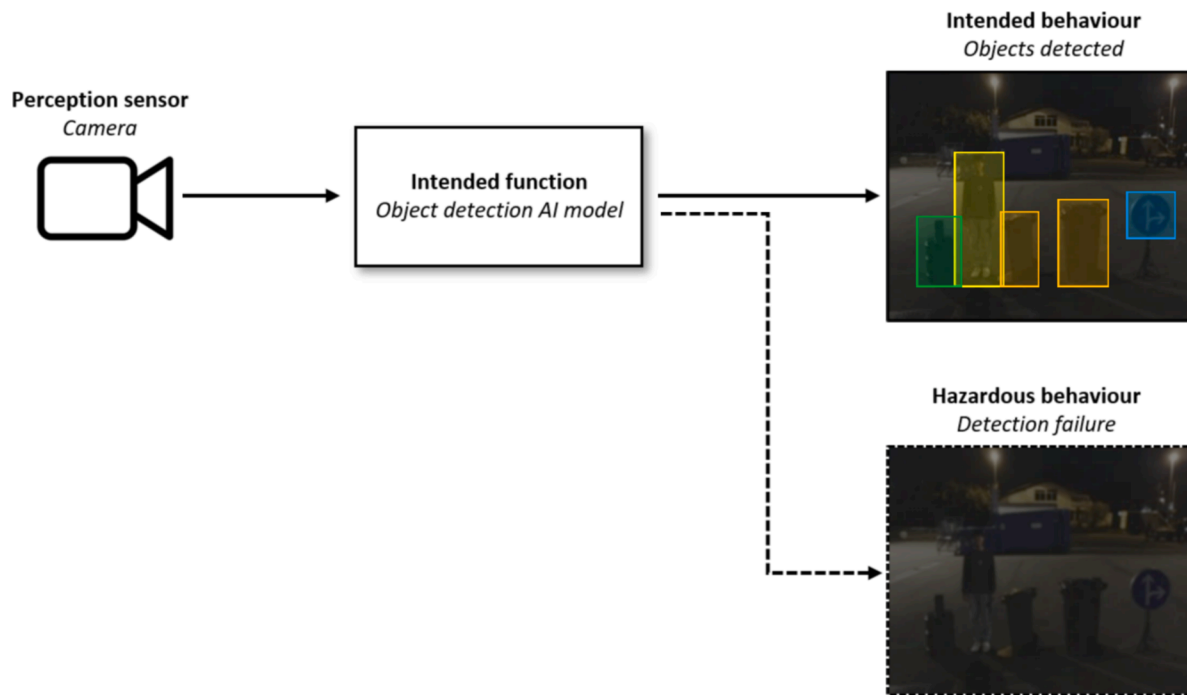


Fig. 1. Example of functional insufficiency of camera-based object detection system. Low-light environments pose a challenge for the system, as the degraded image quality can hinder the detection of important objects (Lee et al., 2023).

conditions due to insufficient illumination, resulting in difficulty capturing clear images of a scene. Consequently, an object detection AI model might struggle to detect or identify objects accurately, even when the camera is functioning properly and without internal faults, due to the insufficient quality of the input runtime image. This can lead to the inability to determine the presence of objects, causing a HAAM to make incorrect decisions and potentially resulting in hazardous situations. In such scenarios, the confidence level in the ability of the camera to perform its intended function may fall below the minimum threshold required for safe operation. To proactively prevent functional deficiencies in the HAAM, it is essential to implement a new mechanism for monitoring the quality of input runtime data.

Image quality plays a critical role in both human vision and computer vision systems. Existing international standards relevant to image quality exclusively focus on aspects that contribute to the fidelity of digital images for human consumption (Board, 2023; Group, 2018; IEC, 2018). These standards inadequately address the specific needs of computer vision applications. Current image quality KPIs, often relying on metrics like sharpness, ISO sensitivity, dynamic range, and colour accuracy, are primarily designed for human perception. However, these KPIs pose significant limitations when applied to diverse computer vision applications (Sebe and Lew, 2013). Even minor image quality degradation, invisible to the human eye, can trigger significant performance drops and even complete failure in a computer vision system. This discrepancy arises from the fundamental difference in how humans and machines extract and interpret information from images (Group, 2018).

Considering a scenario where a computer vision system is deployed for object detection under low-light conditions. While human observers might struggle to discern the objects clearly, the computer vision system, trained on specific features and patterns, could potentially still perform with high accuracy. Conversely, an image with high resolution and vivid colours might be deemed visually appealing to humans, but the presence of noise or artefacts could significantly hinder the ability of a computer vision system to accurately process the information within the image (Sebe and Lew, 2013).

Safety-critical systems increasingly rely on AI/ML models. However,

existing techniques struggle to handle the inherent uncertainty of these models. While model accuracy is commonly measured, it doesn't fully capture the uncertainty in specific outcomes. Kläs et al. (2018) investigates three sources of uncertainty in the application of AI/ML models:

- 1) **Model fit:** This uncertainty arises from the inherent limitations of AI/ML models, which are only approximations of the actual relationship between input and output;
- 2) **Data quality:** The quality of the data used for training and testing the model can significantly impact its performance. Real-world data often suffers from inconsistencies and errors, affecting the model's ability to generalise to unseen scenarios;
- 3) **Scope compliance:** Uncertainty arises when a model is applied outside the specific context for which it was trained and tested. This can happen if the model encounters situations not represented in the training data or if it is used in a different environment than the intended scope.

Uncertainty stems from the inherent limitations and potential issues affecting the accuracy of real-world data used to train and apply the model. This includes limitations in the cleaned data used for training and testing, as well as deviations in the quality and intended scope of the data encountered during actual use.

Several studies have investigated the impact of image quality on AI/ML models for computer vision tasks (Aqqa et al., 2019; Borel-Donohue and Young, 2019; Chen et al., 2017; Dodge and Karam, 2016; Kong et al., 2019). Dodge et al. (2016) found that various image distortions, including blur, noise, and compression artefacts, significantly affect the accuracy of different image classification models. Similarly, Borel-Donohue et al. (2019) observed a decrease in object detection accuracy for YOLO (You Only Look Once) frameworks in unmanned aerial vehicles when encountering degradations like blur and reduced resolution. Aqqa et al. (2019) also reported susceptibility to compression artefacts in image classification models, highlighting the need for improved robustness. These findings emphasise the importance of considering image quality during model development and deployment, with

suggestions for incorporating training with diverse distortions as a potential solution. [Chen et al. \(2017\)](#) address this challenge by incorporating distorted images into the training process via data augmentation and utilising an image quality assessment-based label smoothing technique. This approach improves the model's ability to learn robust features and handle real-world image variations, improving performance across diverse distortion types and levels.

[Kong et al. \(2019\)](#) addresses the challenge of accurately predicting image quality for object detection tasks in diverse real-world scenarios. Existing image quality measures, such as peak-signal-to-noise ratio (PSNR) and structured similarity indexing method (SSIM), primarily focus on human perception and fail to capture the specific needs of object detection algorithms. The authors propose a novel blind image quality model that utilises local and global features (e.g., edge information and estimated object size) to predict the performance of object detection algorithms on an image. This model is trained on a dataset encompassing various distortions (noise, blur, down-sampling) and diverse scene characteristics. Experimental results demonstrate the effectiveness of the proposed model, achieving significantly higher correlations with actual object detection performance compared to traditional image quality measures. This work highlights the importance of considering task-specific quality metrics for automated analysis applications to evaluate image quality.

[Li et al. \(2019\)](#) investigates a deep learning approach for image quality classification in surveillance applications, such as colour casting, blur, signal loss, luminance anomaly, etc. The proposed method utilises a pre-trained InceptionV3 convolutional neural network (CNN) as a feature extractor and a support vector machine (SVM) as a classifier. Training the model with a data-augmented dataset, including various distortion types, significantly improves accuracy compared to using the original dataset. The InceptionV3 model, compared to other tested CNN architectures, extracts features that best describe the surveillance image dataset. Features extracted from the middle layers of the InceptionV3 model perform better than those from higher layers, which may capture less relevant image quality information. The proposed method based on InceptionV3 and SVM demonstrates high classification accuracy and low miss and wrong diagnosis rates for image quality classification in surveillance tasks.

Existing research on image quality primarily focuses on its post-analysis impact on the performance of AI and ML models. However, there is a critical gap in understanding how image quality can affect the functional insufficiency of these models in real time. This insufficiency poses significant risks, especially in safety-critical applications where incorrect or unreliable model outputs can have severe consequences.

This article aims to develop an **Image Quality Safety Model (IQSM)** that can estimate the confidence level in the safety of the intended functionality. The IQSM can help identify and address image quality issues before they negatively impact the performance of an AI/ML-based intended function. It can also provide valuable insights for safety-critical applications, helping to ensure that the AI/ML model is used responsibly and within their limitations.

Earlier work ([Lee et al., 2023](#)) introduced the design and verification for implementing a functionally safe and reliable perception in a generalised manner. Risk parameters, including environmental, obstacle-related, and machine-related parameters, were identified as key aspects that affect the detection reliability of perception sensors/systems. These aspects can potentially cause perception sensors/systems to malfunction, lowering their confidence level. These hazards can be triggered by specific conditions of a scenario, defined as triggering conditions ([ISO, 2022](#)). This article considers these risk parameters as potential triggering conditions for a perception system in HAAM.

This article is organised as follows: first, normative references for the safety of highly automated mobile machines and vehicles are introduced, followed by an explanation of the basic principles of a perception system for HAAM. Then, the article describes the concept and architecture of an IQSM for estimating the confidence level of the safety of the

Table 1

Overview of the referred ISO standards for perception system safety.

ISO standards	Description	Specific target
ISO 18497	Safety of highly automated agricultural machines – principle for design	Highly automated agricultural machines
ISO 21448	Safety of the intended functionality	Road vehicles
ISO/IEC TR 5469	Functional safety and AI systems	General safety-related AI systems
ISO 25119	Safety-related parts of control systems	Tractors and machinery for agriculture and forestry

intended functionality. The dataset generation and training method used for developing the IQSM are presented. Finally, the trained IQSM model is evaluated based on foreseeable hazardous scenarios.

2. Materials and methods

A perception system in HAAM is a complex system that relies on a sophisticated interplay of sensors, advanced algorithms, and electrical/electronic/programmable electronic (E/E/PE) systems to perform their tasks. As the number of automated functionalities in HAAM continues to grow, so does the need for robust safety measures. Ensuring the safety of the HAAM is paramount for the highly automated machine industry.

This part covers safety-related ISO standards necessary for the design of a safe and reliable perception system and the development process of the IQSM for the safety of the intended functionality.

2.1. Standards as a framework

This section introduces four safety-related ISO standards ([Table 1](#)) ([ISO, 2018a, b, 2022, 2024](#); [Tiusanen et al., 2020](#)). These standards provide fundamental guidelines for designing AI/ML-enabled E/E/PE perception systems to ensure safety and reliability in highly automated machines and vehicles and address potential risks associated with AI/ML-based technologies in automated applications. While ISO 21448 (SOTIF) was primarily designed for road vehicles, its underlying principles and methodologies can be extended to the development of HAAMs.

ISO 25119 series sets out general principles for designing and developing safety-related parts of control systems on tractors used in agricultural machines ([ISO, 2018b](#)). The series defines functional safety as the absence of unreasonable risk due to hazards caused by the malfunctioning behaviour of E/E/PE safety-related systems. ISO 25119 series primarily addresses the properties and measures necessary to mitigate risks and ensure the reliability of E/E/PE safety-related systems. The reliability of a perception system must be subject to an assessment according to the ISO 25119 series or an equivalent functional safety standard (e.g. ISO 13849 or IEC 61508).

However, in the context of HAAM, E/E/PE system faults are not the sole safety-related factor. Even if E/E/PE systems function flawlessly, the intended functionality or implementation can lead to hazardous situations, as seen in the previous example ([Fig. 1](#)). There, the camera might operate without any E/E/PE faults, yet the object detection system could still generate unreliable results due to insufficient image quality, potentially causing an unsafe situation. Such potentially hazardous behaviour can arise from the inability of the function to correctly perceive the environment, the lack of robustness of the function, system, or algorithm with respect to sensor input variations, fusion heuristics, diverse environmental conditions, or unexpected behaviour caused by the decision-making algorithm and/or differing human expectations.

The absence of unreasonable risk resulting from hazardous behaviour related to functional insufficiencies is defined as the safety of the intended functionality (SOTIF). Functional safety addressed by the ISO 25119 series and the SOTIF are complementary aspects of safety.

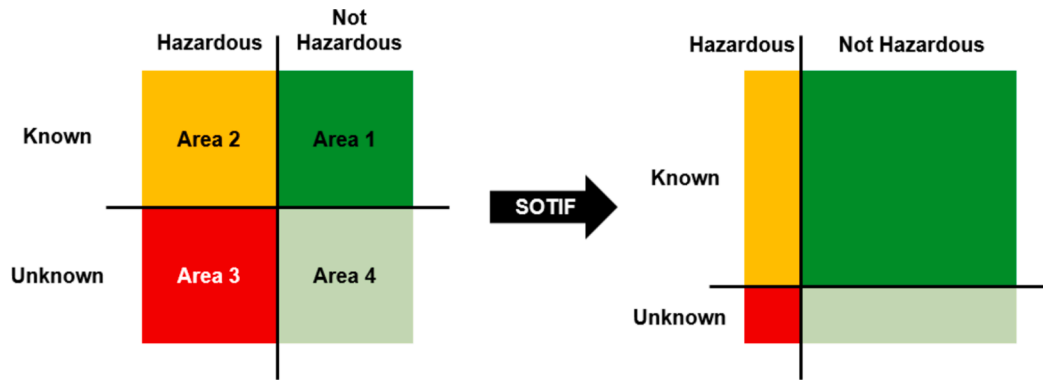


Fig. 2. Visualisation of scenario categories. Area 1: known, not hazardous scenarios. Area 2: known, hazardous scenarios. Area 3: unknown, hazardous scenarios. Area 4: unknown, not hazardous scenarios.

Table 2

Overview of the cause of hazardous events and their safety issues (ISO, 2018b, 2022).

Cause of hazardous events	Safety issues	Within scope of
E/E/PE system faults	Random hardware faults (e.g., ageing of E/E/PE component) Systematic faults (e.g., software bugs or programming errors) Common cause (e.g., physical separation between signal and power)	ISO 25119 series
Functional insufficiencies Functional insufficiencies of artificial intelligence-based algorithms	Insufficiency of specification (e.g., limited sensor range) Performance limitations (e.g., perception failures under low-light conditions) Environment conditions (e.g., heavy snow, rain, or fog) Foreseeable misuse by persons (e.g., ignoring warnings)	ISO 21448

ISO 21448 covers cases where the intended design itself is insufficient to ensure safety in the absence of E/E/PE system faults. It guides the applicable design, verification and validation measures needed to achieve and maintain the SOTIF (ISO, 2022). ISO 21448 applies to intended functionalities where proper situational awareness is essential to safety and where such situational awareness is derived from complex sensors and processing algorithms. ISO 21448 describes a conceptual abstraction representing a goal of the SOTIF activities (Fig. 2). The scenarios which are part of the relevant use cases are classified into four areas. The goal of SOTIF activities is to evaluate the potentially hazardous behaviour in areas 2 and 3 and to argue that the residual risk caused by these scenarios is sufficiently low. It is expected that the residual risk due to areas 2 and 3 will be reduced. The confidence in the achievement of the SOTIF will be increased by the growing scenario set in Area 1.

Functional safety and SOTIF are complementary aspects of safety. Table 2 illustrates how the possible causes of hazardous events map to existing standards. ISO 25119 addresses E/E/PE system faults, such as random hardware, systematic, and common cause faults, that can cause hazards due to deviations from the intended functionality. SOTIF-related hazards, such as the inability of the function to correctly perceive the environment; the lack of robustness of the function, system, or algorithm (including those using AI or ML) with respect to sensor input variations, heuristics used for fusion, or diverse environmental conditions; the unexpected behaviour due to decision-making algorithm and/or divergent human expectations, are covered by ISO 21448.

AI technology is often used in cases where physical phenomena are so complex or at such a small scale or unobservable without influencing

Table 3

Overview of risk factors in AI safety-related systems (ISO, 2024).

Risk factors	Reasons
Transparency and explainability	Decisions made by AI models, especially by models of high complexity or models derived from machine learning algorithms, are more difficult to understand for humans.
Model uncertainty	Even though AI systems are often based on deterministic models, their outputs can still be probabilistic. This is because AI systems are trained on a lot of data, but the data may not be representative of all possible situations. As a result, AI systems can make mistakes or behave unexpectedly in new or unexpected situations.
Data drift	The distribution of runtime input data deviates from the distribution of data used in the training phase, resulting in performance degradation. This discrepancy can be caused by external factors such as different lighting conditions in training and previously unseen data.
Concept drift	The relationship between input variables and model output changes over time, which can lead to inaccurate predictions and poor performance.
Adversarial and intentional malicious inputs	Inputs intentionally crafted for possible misbehaviour lead AI models to compute incorrect outputs without destroying the integrity of software execution (e.g., buffer overflow or integer overflow) or causing malfunctions at the software level.
Random AI hardware faults	Hardware faults can disrupt the intended execution of an algorithm by interfering with its control flow, corrupting data inputs, and directly altering output values, leading to erroneous results.

the experimental data that, consequently, there is no scientific model of the underlying behaviour. In these cases, the model of the AI technology is possibly neither transparent nor complete. The compatibility of the model with functional safety requirements is hard to demonstrate (ISO, 2024). Therefore, when AI technology is used, especially in safety-critical systems, special considerations must be considered (ISO, 2022).

The use of AI technology and AI systems is currently not treated in existing functional safety standards, such as the ISO 25119 series. ISO/IEC TR 5469 focuses on the aspects of safety functions performed by a safety-related system by using AI technology, either within the safety-related system or during the design and development of the safety-related system. It describes a framework for applying functional safety principles to AI systems, providing a valuable starting point for developing safe AI systems in safety-critical applications. This standard introduces the properties, related risk factors, available methods and processes relating to 1) use of AI inside a safety-related function to realise the functionality; 2) use of non-AI safety related functions to ensure safety for AI controlled equipment; 3) use of AI systems to design and develop safety related functions.

Table 3 shows related functional safety risk factors of AI systems.

Table 4

Summary of risk parameters for the safety and reliability of perception systems (ISO, 2018a, 2022, 2024; Lee et al., 2023).

Risk parameters	Cause of hazardous events	Within scope of
Environmental parameters	<i>Illumination conditions (e.g., sun glare, darkness)</i> <i>Atmospheric precipitation and conditions (e.g., rain, fog, snow)</i> <i>Interferences (e.g., electromagnetic interference, RADAR, LiDAR)</i> <i>Irregularities (e.g., terrain irregularities, crop irregularities)</i>	ISO 18497
Obstacle-related parameters	<i>Obstacle physical properties (e.g., poor reflection properties, small size of obstacles)</i> <i>Obstacle states (e.g., fast speed of obstacles)</i>	
Machine-related parameters	<i>Mechanical disturbance (e.g., vibration, stress)</i> <i>Durability, wear, and ageing (e.g., camera lens becomes dull, sensor misalignment)</i> <i>Obscuration on sensor (e.g., raindrops on the lens)</i> <i>Machine states (e.g., high operating speed)</i>	
SOTIF-related parameters	<i>Insufficiency of specification (e.g., limited sensor range)</i> <i>Performance limitations (e.g., perception failures under low-light conditions)</i> <i>Environment conditions (e.g., heavy snow, rain, or fog)</i> <i>Foreseeable misuse by persons (e.g., ignoring warnings)</i>	ISO 21448
AI-related parameters	<i>Transparency and explainability (e.g., black box complex AI model)</i> <i>Complexity of the environment (e.g., model uncertainty)</i> <i>Environmental changes (e.g., data drift, concept drift)</i> <i>Adversarial and intentional malicious inputs (e.g., AI model attacks)</i> <i>AI hardware issues (e.g. memory-based errors)</i>	ISO/IEC TR 5469
E/E/PE component-related parameters	<i>Random hardware faults (e.g., ageing of E/E/PE component)</i> <i>Systematic faults (e.g., software bugs or programming errors)</i> <i>Common cause (e.g., physical separation between signal and power)</i>	ISO 25119 series

ISO 18497 specifies principles for the design of highly automated agricultural machine operations to achieve safe operation (ISO, 2018a). This document deals with the significant hazards, hazardous situations, and events relevant to agricultural machinery and tractors that allow highly automated field operations. It introduces fundamental requirements for the design of a perception system.

According to ISO 18497, HAAM must be provided with a perception system capable of detecting objects, people, and obstacles in its path under foreseeable operating conditions to ensure safe operation. The perception system relies on perception sensors, positioning/guidance systems, associated algorithms required for the identification and classification of significant objects relative to the HAAM and the safety-related control system for controlling the machine without the need for human intervention (ISO, 2017, 2018a, 2019).

To ensure the safety of perception systems, faults resulting from reliability issues, such as a result of a hardware failure in the perception system, as well as malfunctions caused by conceptual or software issues, must not lead to a dangerous condition, regardless of whether the perception system is functioning as intended under a fault-free condition in a variety of environmental parameters. In the design of perception systems, it is crucial to consider not only the safety of E/E/PE components but also the objects to be detected, the foreseeable obscurity and the possibility of sensor obscurity due to operational conditions.

A previous study (Lee et al., 2022; Lee et al., 2023) identified risk parameters based on ISO 18497. These parameters included environmental, obstacle-related, and machine-related factors that affect the performance of perception sensors/systems. However, the study did not explicitly address the challenges posed by AI systems. Additional risk parameters have been incorporated to comprehensively assess and mitigate the potential risks posed by AI systems based on ISO/IEC TR 5469 to address these gaps. Functional safety and SOTIF-related risks are also incorporated, providing a more holistic risk assessment framework.

Table 4 provides a summary of the risk parameters that influence the safety and reliability of perception sensors/systems. These parameters can lead to hazardous behaviour and safety hazards for the machine and its surroundings, such as failure to detect an obstacle or false detection of non-existent obstacles.

2.2. Concept of image quality safety model

The concept of an **Image Quality Safety Model (IQSM)** is to estimate the confidence level that a runtime input image is sufficient to safely perform its intended function in real time (Fig. 3). The IQSM provides an output of confidence level. The provided confidence level is in the form of a percentage chance that estimates the safety of its intended functionality (ISO, 2022). If the confidence level falls below the “minimum performance threshold” required for safe operation, the IQSM estimates that the intended function is in an “unsafe” state (ISO, 2018a).

YOLO v8 is a popular object detection algorithm that has gained immense popularity due to its remarkable accuracy and speed (Reis et al., 2023). It is a highly efficient algorithm that can process and detect objects in real time, making it suitable for perception systems in HAAMs. In this article, YOLO v8 is used as an example of an intended function. It was pre-trained on the COCO dataset, which has 80 labels (Lin et al., 2014).

The following situations were identified as hazardous scenarios for cameras, and the following scenarios were used for the validation of the IQSM as safety requirements (Lee et al., 2023):

- Raindrops on the lens can cause distortion in the image and make it difficult to distinguish objects.
- Dust or other obscuring on the sensors itself can reduce the sensor field of view.
- Obstacles are occluded due to crops, dust, fog, snow, rain or other obscuring.

2.3. Development of image quality safety model

This section details the entire development process of the IQSM. Fig. 4 illustrates the steps involved in developing an IQSM. The process consists of four major phases: data preprocessing (Section 2.3.1), labelling (Section 2.3.2), training (Section 2.3.3), and test (Section 2.3.4 and 3).

2.3.1. Data preprocessing phase

An initial dataset is a subset of the data distribution used to train the intended function (i.e., the subset of the COCO dataset used for training the YOLO v8 model). The initial dataset contains colour images containing at least one significant object. An additional augmented dataset was created to generate a wide variety of low-quality images in pre-defined hazardous scenarios. This dataset was generated by adding modified copies of existing images or newly created synthetic images from the initial dataset (Fig. 5). The training dataset for IQSM comprises the initial and low-quality augmented datasets.

Pixel density, contrast, noise, and blurriness are key metrics of image quality that significantly affect the performance of object classification AI models (Aqqa et al., 2019; Borel-Donohue and Young, 2019; Chen et al., 2017; Dodge and Karam, 2016; Kong et al., 2019; Li and Liu,

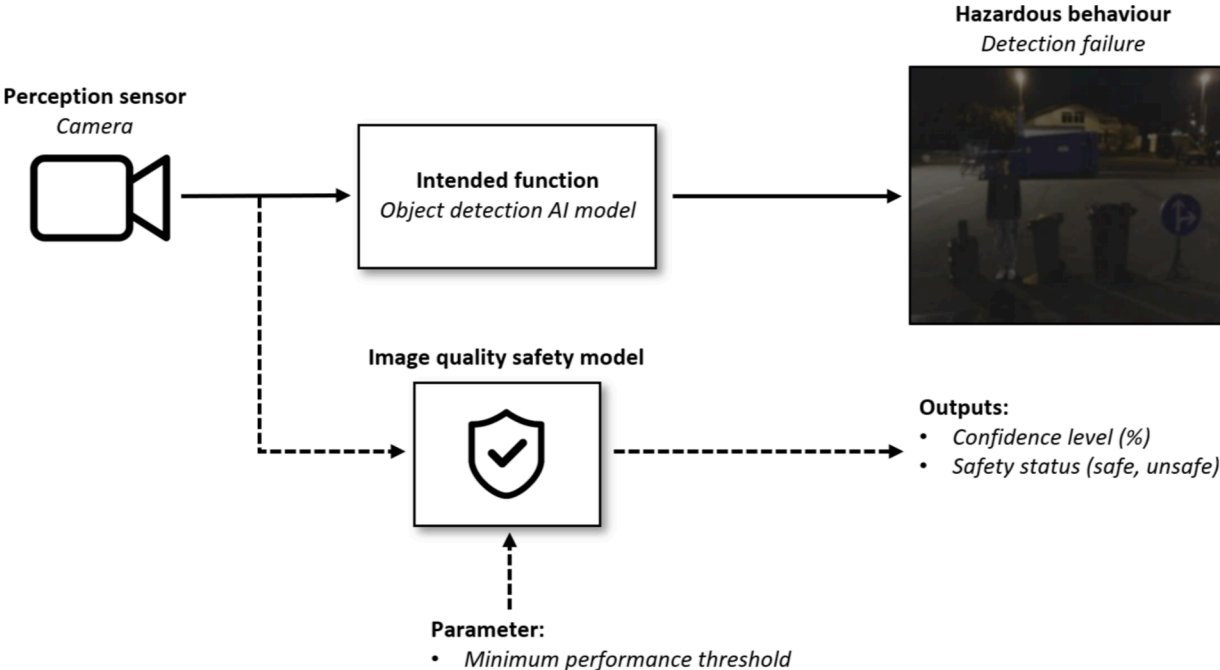


Fig. 3. Concept of an Image Quality Safety Model.

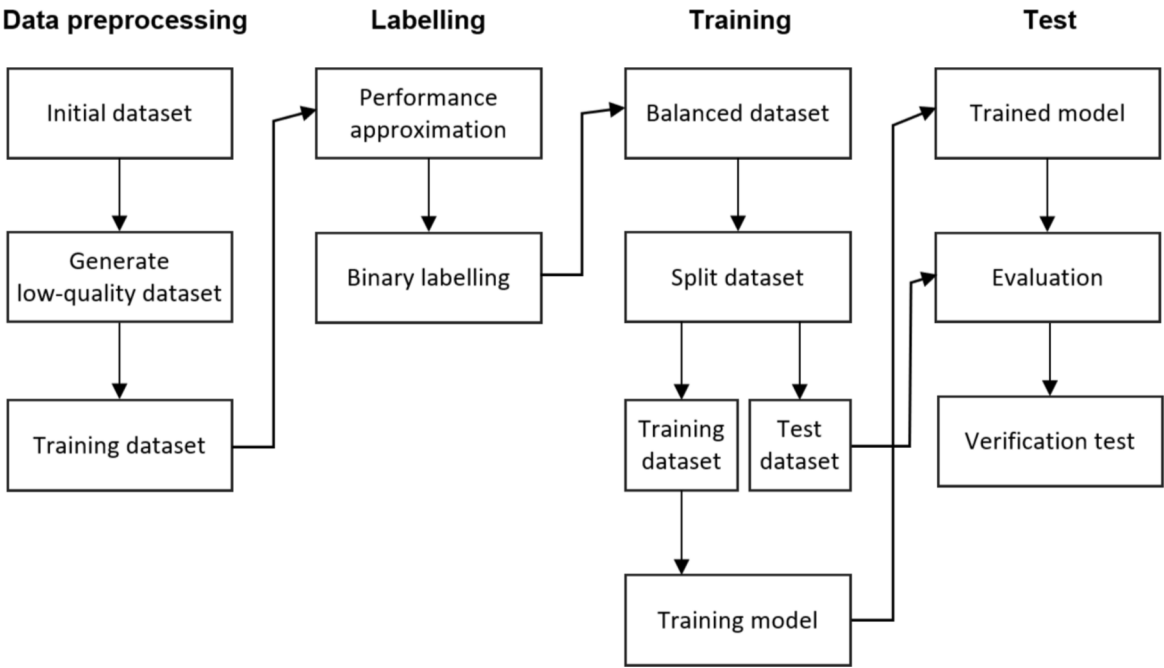


Fig. 4. Overview of IQSM training procedure.

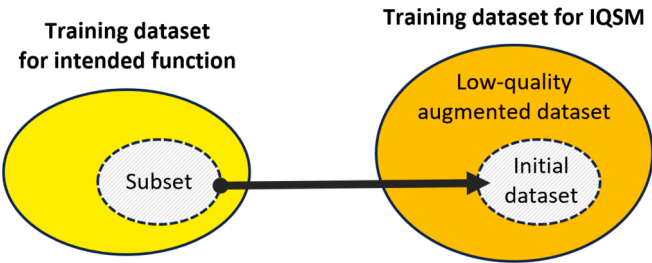


Fig. 5. Procedure for generating training dataset.

2019). Filters related to these key metrics were specifically considered to generate low-quality images, which are of such a tough quality that object classification AI models are barely able to detect objects in images. In addition to the filters related to the key metrics, additional image filters relevant to the predefined hazardous scenarios were also used to generate synthetic images.

1) Gaussian noise

Gaussian noise is a random signal with a normal distribution, meaning that its values are distributed around a mean value with a



Fig. 6. Original image (left), noisy image (middle), and blurred image (right).

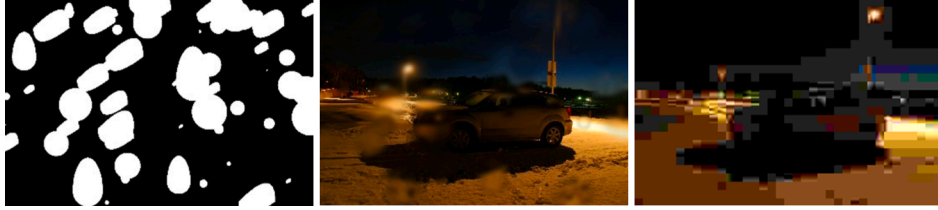


Fig. 7. Mask image (left), local blurred image (middle), and lossy compressed image (right).

specific variance. This type of noise is often found in many natural systems and can be added to signals in various fields, such as communication, electronics, and image processing. In this context, a Gaussian noise filter was applied to create noisy images from the initial dataset (Fig. 6 – middle). To effectively impact the performance of the intended function in the images, the range of variance was experimentally determined, and the variance was randomly selected, while the mean was set to zero.

$$N_{noise}(i, j) \sim N(\mu, \sigma^2) \quad (1)$$

$$I_{noise}(i, j) = I(i, j) + N_{noise}(i, j) \quad (2)$$

where

$N_{noise}(i, j)$ is the Gaussian noisy value at the pixel (i, j) , generated using the Gaussian distribution

μ is the mean and σ^2 is the variance of the Gaussian distribution

$N(\mu, \sigma^2)$ denotes the Gaussian distribution

$I_{noise}(i, j)$ is the intensity value of the noisy image at pixel (i, j) .

$I(i, j)$ is the intensity value of initial image at pixel (i, j) .

2) Gaussian blur

Gaussian blur applies a mathematical function known as a Gaussian distribution to each pixel in an image. This process smooths out pixel values and reduces the contrast between adjacent pixels. In this context, Gaussian blur filter (Fig. 6 – right) was used to generate images with reduced or distorted detail. The degree of blur applied to an image is determined by the standard deviation of the Gaussian distribution. The standard deviation in both vertical and horizontal directions was randomly chosen, providing a sufficient amount to impact the performance of the intended function in the images.

$$G(i, j) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{i^2 + j^2}{2\sigma^2}\right) \quad (3)$$

$$I_{blur}(i, j) = \sum_{u=-k}^k \sum_{v=-k}^k G(i, j) I(i + u, j + v) \quad (4)$$

where

$G(i, j)$ is Gaussian kernel value at position (i, j) within the kernel matrix.

σ is the standard deviation of the Gaussian distribution

$I_{blur}(i, j)$ is the intensity value of the blurred image at pixel (i, j) .

$I(i, j)$ is the intensity value of initial image at pixel (i, j) .

3) Local blur

An artificial raindrop generation algorithm was used to generate realistic and diverse artificial raindrops in images (Fig. 7 – left and middle) (Soboleva and Shipitko, 2021). The algorithm allows the generation of raindrops in the shapes of a circle, an egg, and a combination of two bezier curves. The radius, coordinates of the drop centre and shape are randomly selected for each drop. Then, it creates transparency and background image masks based on the chosen shape and brightness. Finally, it blends the masked background and a blurred portion of the original image to create the final raindrop with darker edges, mimicking natural raindrops. The number and radius of raindrops were subjectively chosen.

4) Lossy compression

Pixel density refers to the number of pixels per unit area in an image.

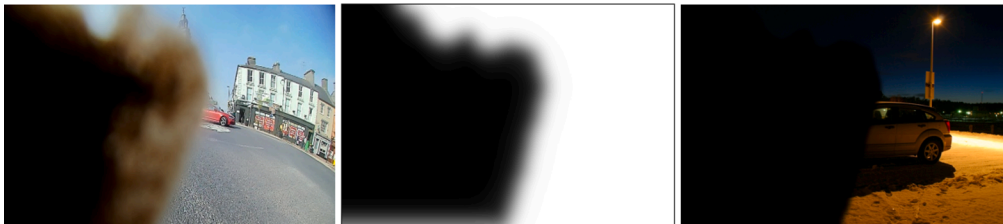


Fig. 8. Mask filtering process to generate images that mimic dirt on lens. The image on the left shows a real contamination scenario. The middle image shows the mask filter in a form like the left image. The right image shows overlaid onto a new image.

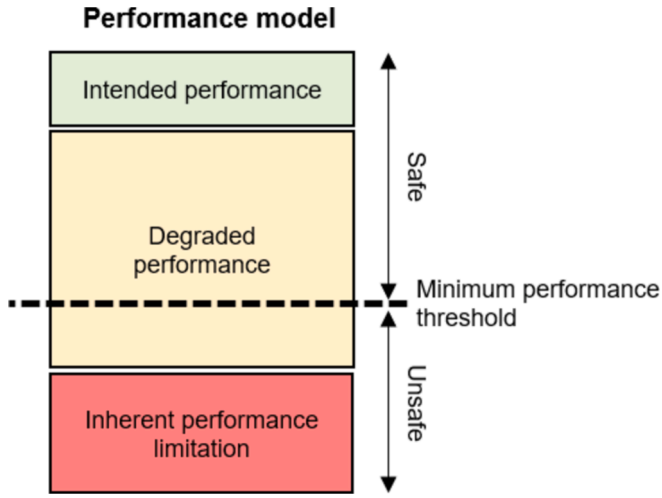


Fig. 9. Performance model of an intended function.

Higher pixel density can improve image quality by providing finer details. Image compression techniques reduce the file size by removing redundant information, but excessive compression can lead to loss of image quality and negatively affect object detection performance due to loss of detail and artefacts. Lossy Compression methods reduce the pixel density of images by discarding some image data. It achieves higher compression ratios compared to lossless compression but may result in a slight loss of image quality (Aqqa et al., 2019). To introduce variations in quality, the initial images were compressed using JPEG, a lossy compression algorithm (Fig. 7 – right).

5) Mask filter

Different patterns of mask filters were used to generate images that mimic dirt on lens (Fig. 8). The mask-only images with different patterns

were collected from authentic contamination images, open-source datasets, or software generated randomly. The edges of the mask-only images were smoothed using Gaussian blur to closely mimic the actual contamination effect. The standard deviation was randomly chosen. The blurred mask-only images are overlaid onto the initial images, and the pixel values in the masked areas are modified, while the pixel values in the unmasked areas remain unchanged. Masked images are generated by the convolution of a blurred mask filter image to a clear image.

$$M_{blur}(i, j) = M(i, j) * G(i, j) \quad (5)$$

$$I_{mask}(i, j) = I(i, j) * M_{blur}(i, j) \quad (6)$$

where

$M_{blur}(i, j)$ is the intensity value of the blurred mask-only image at pixel (i, j) .

$M(i, j)$ is the intensity value of the mask-only image at pixel (i, j) .

$G(i, j)$ is Gaussian kernel value at position (i, j) within the kernel matrix.

$I_{mask}(i, j)$ is the intensity value of the masked image at pixel (i, j) .

$I(i, j)$ is the intensity value of the initial image at pixel (i, j) .

2.3.2. Labelling phase

In this article, the performance of the intended function (or algorithmic performance) is defined into three levels (Fig. 9):

- **Intended performance** represents the optimal level of performance that the intended function can achieve in designated application contexts, such as when the distribution of runtime input data adheres to the training data distribution.
- **Degraded performance** refers to performance that is lower than the intended performance. This arises when the data quality is degraded by external factors, such as seasonal change, or when the runtime input data deviates from the training data distribution.
- **Inherent performance limitation** refers to performance that is unable to achieve the intended function due to constraints within the

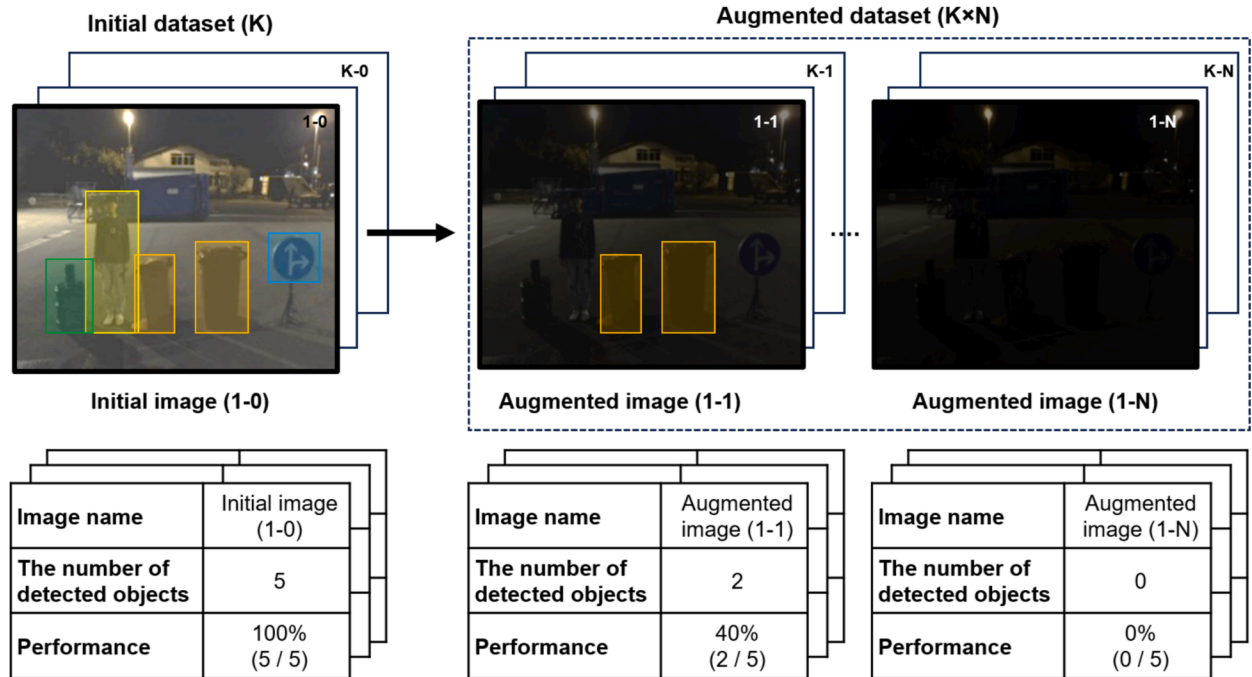


Fig. 10. Example of performance approximation method. The performance of augmented images is calculated by the ratio of the number of detected objects in each augmented image to the number of detected objects in the initial image. “K” denotes the total number of initial images in the initial dataset. “N” denotes the total number of used low-quality filters. The nomenclature “K-0” refers to a Kth initial image in the initial dataset. The nomenclature “K-N” refers to the filtered Kth image using the Nth filter.

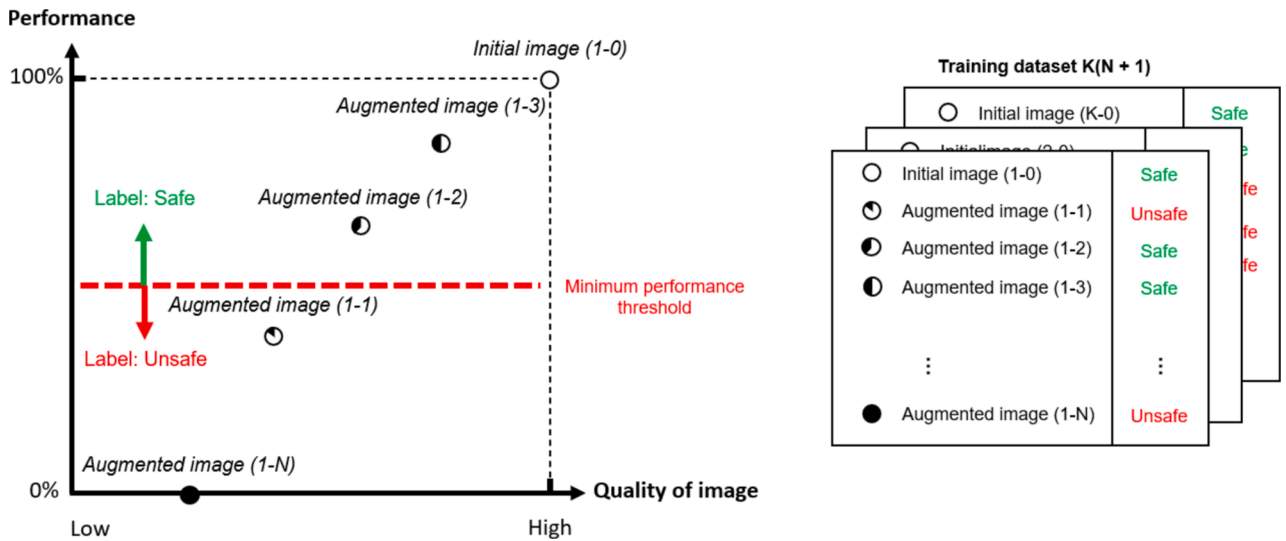


Fig. 11. Example of a binary labelling method. “K” denotes the total number of initial images in the initial dataset. “N” denotes the total number of used low-quality filters.

training data distribution (e.g., data distribution does not reflect the actual distribution in the application contexts) or limitations in the model architecture or the underlying technology.

The performance is in the form of a percentage chance, with higher percentages indicating better performance. The selection of a “minimum performance threshold” for a perception system is determined by the safety requirements associated with its intended application. The minimum performance threshold can be determined through statistical analysis of the performance of an intended function. This article sets an arbitrary performance requirement for the intended function as “the performance of the intended function must not fall below 50 %”. If the performance falls below 50 %, the perception system is insufficient to

ensure the safety of the intended operations under triggering conditions.

The performance corresponding to each training image was approximated to train the IQSM on the relationship between image quality and the performance of the intended function. This article uses only the number of detected objects in images, regardless of whether they are correctly classified or not, as a metric to approximate the runtime performance of the intended function. (Fig. 10). Since the performance of the intended function already achieved high classification accuracy when evaluated on the COCO dataset (Lin et al., 2014), the prediction results on the initial dataset, which is a subset of the training data distribution of the intended function, can be assumed to be correctly classified for simplicity. Therefore, the performance of the intended function on this initial dataset is considered 100 %. The

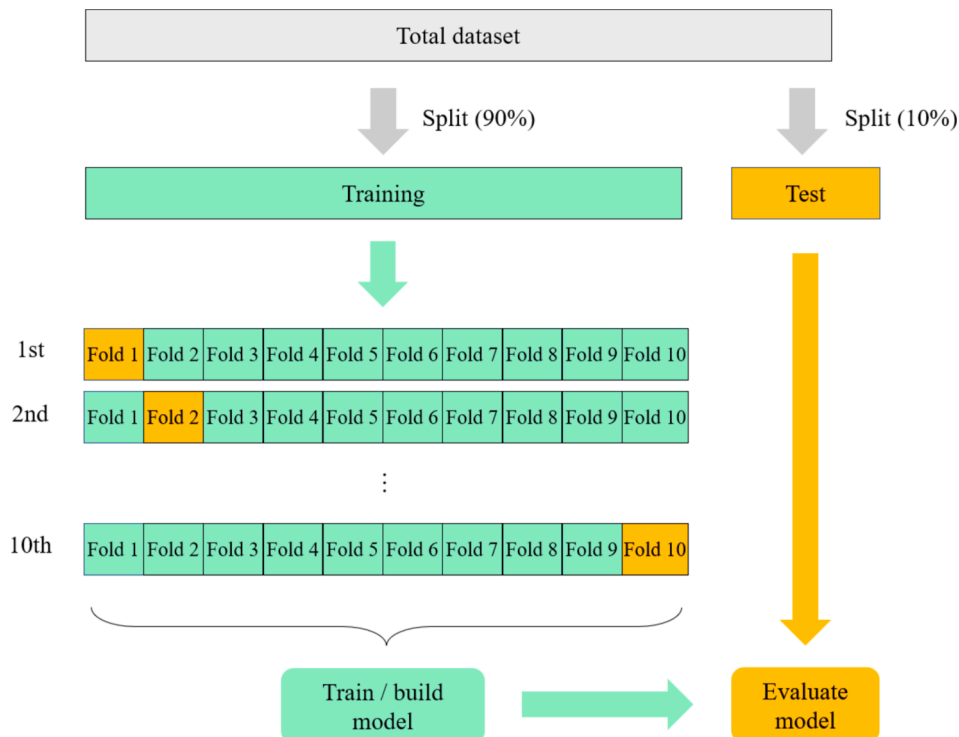


Fig. 12. Overview of a k-fold cross-validation process.

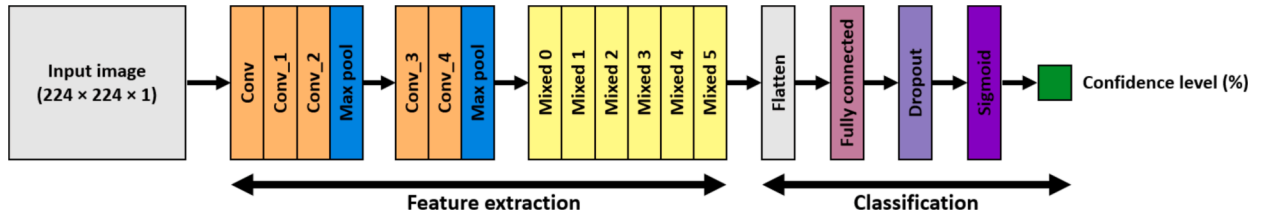


Fig. 13. The architecture of an IQSM in Keras API (Gulli et al., 2019).

performance of the intended function on each augmented image, generated from an initial image, was approximated by the ratio of the number of detected objects in the augmented image to the number of detected objects in the initial image (Fig. 10).

Images in the training dataset, in both initial and augmented datasets, were labelled as “safe” if their performance exceeded the defined minimum performance threshold and as “unsafe” if their performance fell below the threshold (Fig. 11). The IQSM can learn the quality of each image through the approximated performance of the intended function. This enables the IQSM to estimate the performance of the intended function within the context of the trained quality data distribution.

2.3.3. Training phase

The exact number of images was randomly chosen from each label to address an imbalance in the training dataset. This balanced dataset was then used for training, validation, and testing. The images of varying dimensions were resized to 224 x 224 pixels and converted to greyscale images. The balanced dataset was split into 90 % for training and 10 % for testing, and to ensure the model’s robustness and generalizability, 10-fold cross-validation was employed (Fig. 12) (Berrar, 2019). The training dataset was randomly divided into ten equal-sized folds, each containing 10 % of the data for validation and the remaining 90 % for training. The model was trained on the training set during each fold and evaluated on the validation set.

The final performance metric was calculated by averaging the results from all folds. The IQSM was developed and implemented using the TensorFlow deep learning library with the Keras API (Gulli et al., 2019). The model was trained using the binary cross-entropy loss function to measure the difference between the predicted probability distribution of the model and the true probability distribution of the target. A stochastic gradient descent optimiser was utilised with a learning rate of 0.0001 to minimise the binary cross-entropy loss function by updating the model weights based on the gradients of the loss function with respect to the weights. Early stopping was employed to prevent the model from overfitting or underfitting, which depended on the model’s validation performance. The training process was terminated if there was no observable improvement in performance for 20 consecutive epochs, and the validation loss function was utilised as the performance metric.

Fig. 13 explains the architecture of the IQSM. The IQSM considers an architecture of the convolutional neural network (CNN) for binary classification. This can include several layers such as convolutional layers, max pooling layers, dropout layers, and fully connected layers. A pre-trained inception V3 model is used as a feature extractor to extract image quality features.

Inception V3, developed by Google (Szegedy et al., 2016), is a deep learning model for image classification, object detection, and segmentation. Inception V3 is a convolutional neural network architecture that uses “inception modules” to improve accuracy and efficiency. The network has several layers of convolutions and pooling, and it uses batch normalisation to speed up training. Mixed5 is one of the layers in the Inception V3 architecture. It is called “mixed” because it combines features from different sizes and types of convolutions. Mixed5 is the fifth mixed layer in the network, located near the top of the architecture. It consists of several inception modules containing multiple convolutional layers with different filter sizes and pooling operations.

This article modified Inception V3 by feeding the output of Mixed5 into new final classification layers. Mixed5 was selected experimentally to leverage a sufficiently large feature map. This adaptation allows the IQSM to train on the overall quality of the environment rather than focusing solely on specific object features. Consequently, the IQSM can estimate the confidence level of its intended function, even in an object-free environment.

The IQSM takes a greyscale image size of 224 × 224 as an input image. Colour images are converted to greyscale images. The input image is passed through pre-trained Inception V3 to extract relevant features from the image. After the feature extraction process, the output from the final convolutional layer is a set of feature maps. The feature maps are flattened into a one-dimensional vector to feed these features into a fully connected layer. Next, the flattened features are passed through a dense layer with a rectified linear unit (ReLU) activation. The dense layer has 32 neurons and can learn complex patterns and relationships between the extracted features. To prevent overfitting, dropout regularisation with a rate of 0.5 can be applied after a dense layer. During training, dropout randomly drops out a fraction of the activations in the previous layer, forcing the model to learn more robust features that generalise well to new data. Finally, the output from the final dense layer is passed through a sigmoid activation function. The sigmoid function maps the input to a value between 0 and 1, which can be interpreted as a probability.

2.3.4. Test phase

1) Model evaluation method

After training was completed, the performance of the IQSM model was evaluated on a test set. The metrics adopted include overall classification accuracy, recall, precision, and F1-score. The metrics are defined as follows (Goutte and Gaussier, 2005):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-score} = \frac{2 \times (\text{Recall} \times \text{Precision})}{\text{Recall} + \text{Precision}}$$

Accuracy is a measure of how often the classifier is correct. Precision is a measure of how often the classifier correctly predicts a positive result. Recall is a measure of how often the classifier correctly identifies positive results out of all the actual positive results. The f1-score is calculated as the harmonic mean of precision and recall. The acronyms TP, TN, FP, and FN refer to true positive, true negative, false positive, and false negative, respectively. In this article, if the high-quality image is correctly classified as safe state, it is counted as TP, while if incorrectly classified as unsafe, it is counted as FN. On the other hand, if a low-quality image is classified correctly as an unsafe state, it is counted as TN, and the incorrectly classified as safe state is FP. A confusion matrix

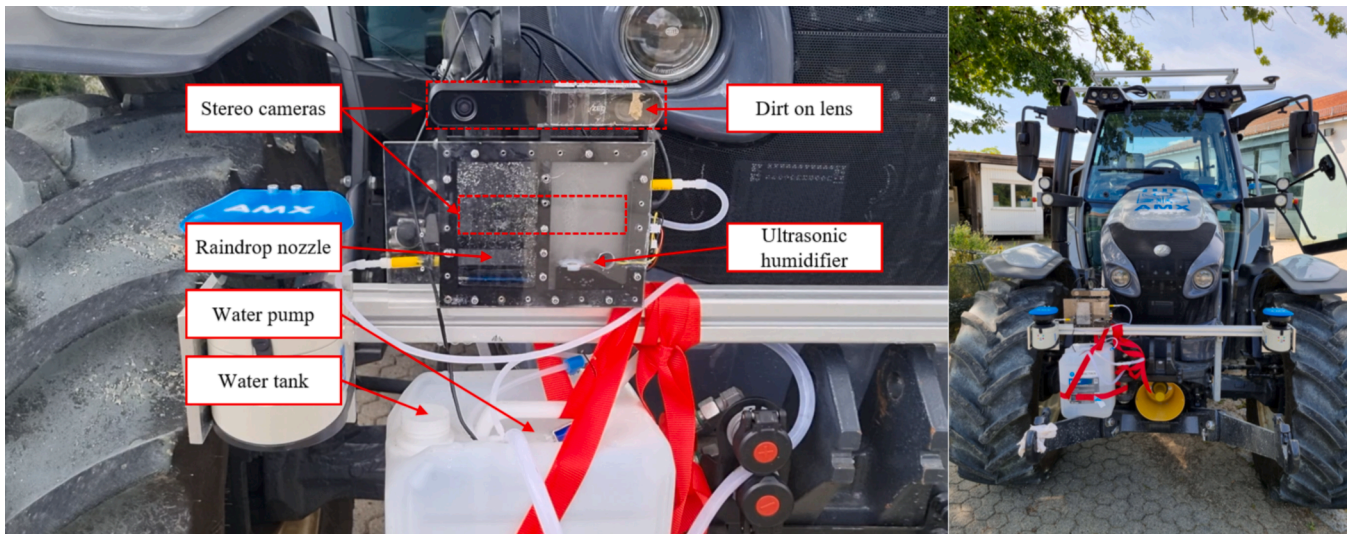


Fig. 14. Low-quality image generator attached to TUM AMX tractor.

was plotted to depict the number of correctly classified images, and a classification report was generated using the scikit-learn metrics function.

2) Verification test method

In this article, a practical test was conducted to see if the trained IQSM behaves as expected and delivers the desired results. YOLO v8, the intended safety function, was used together to confirm whether objects could be detected within the test environment.

As a means for verifying the performance of the IQSM under hazardous scenarios, a low-quality image generator was developed, capable

of generating clear, dirty raindrops on lens and foggy images (Fig. 14). A 3D-printed bi-partite chamber is sandwiched between two acrylic clear panels and placed in front of the two lenses of a stereo camera (Stereo labs, ZED mini) (Porav et al., 2019). The left chamber is filled with water droplets dispersed by an internal raindrop nozzle, while the right chamber contains fog produced by an ultrasonic humidifier. A second stereo camera (Stereo labs, ZED 2i) is mounted on top of the chamber, one of its lenses obscured by dirt, while the other lens is clear. A water pump (Decdeal, submersible pump) continuously circulates water within the enclosed system, replenished from a water tank. The low-quality image generator was mounted in front of a tractor. It captured 1920×1080 pixel colour images using an NVIDIA Jetson Nano

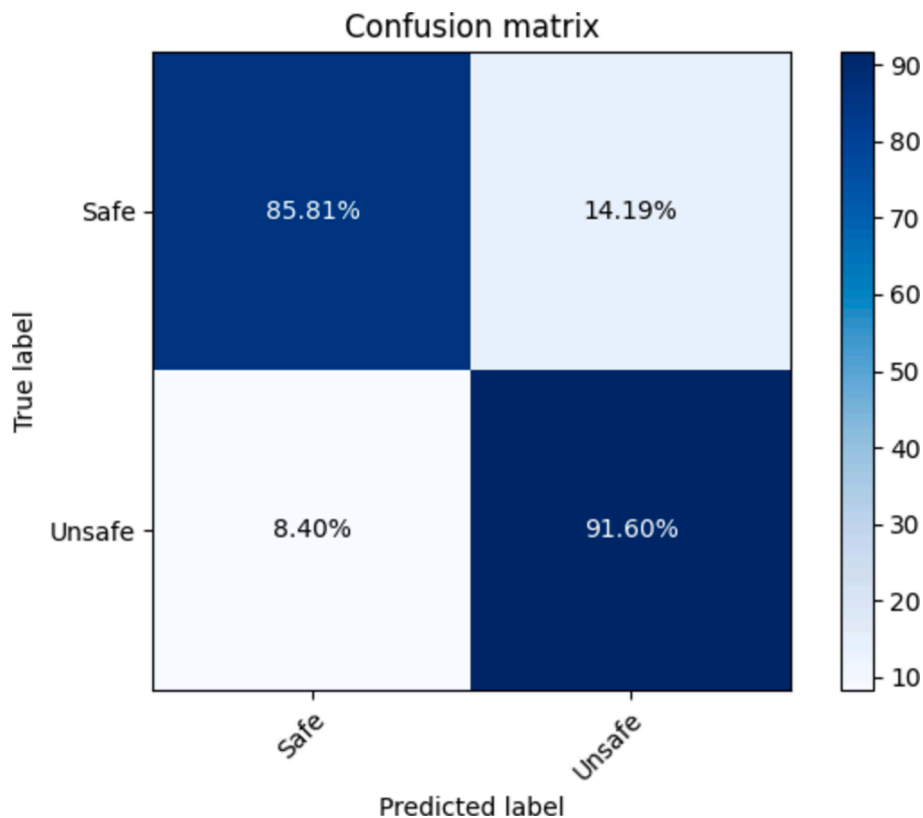


Fig. 15. Confusion matrix for the test dataset.

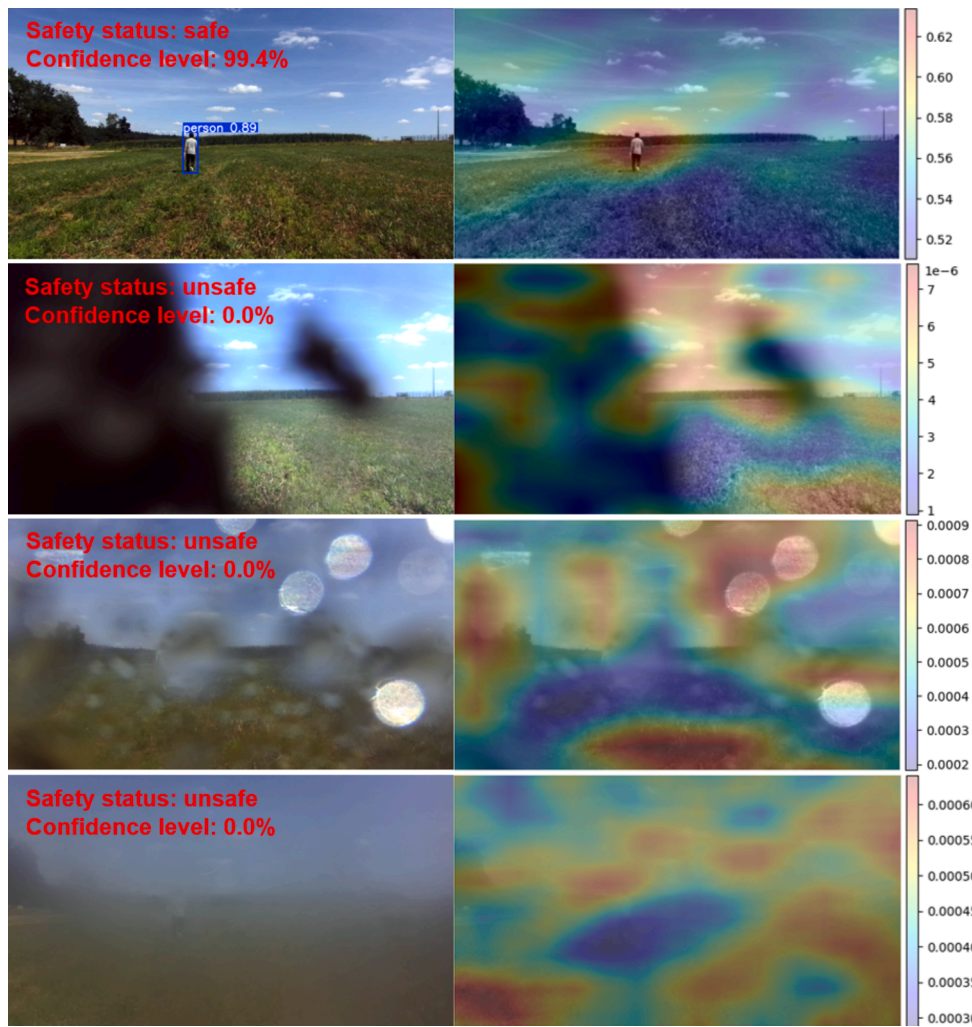


Fig. 16. Scene 1: a grass field scene with clear, dirty, raindrops, foggy conditions and corresponding saliency maps.

embedded computer.

On a bright day, four different synchronised images were captured while driving through a field near the Dürnast research station of Technical University of Munich (TUM). The tractor travelled on unpaved roads and grass at an operating speed of about 7 km/h, encountering test obstacles such as humans and cars.

The interpretability of the decision-making process of the IQSM presents a significant challenge. To address this, randomised input sampling for explanation (RISE) was employed (Petsiuk, 2018). The RISE operates on AI/ML models. It estimates the importance of input features for the prediction of the model by systematically probing the model with masked versions of the input and analysing the corresponding outputs. RISE generates a **saliency map**. This map is constructed as a weighted sum of the random masks, where the weights correspond to the probability scores assigned by the model to the masked inputs, adjusted to account for the random masks' distribution. This approach effectively highlights the regions within the input image that exert the most significant influence on the prediction of the model.

3. Results and discussion

In this article, 33,228 images were used to train the IQSM and 3,322 images were tested to evaluate the trained IQSM. With 20 epochs, the IQSM achieved validation accuracy of about 90 %. This indicates that this model was able to learn the distinctive feature of image quality. Fig. 15 summarises the performance of the IQSM by comparing

predicted labels to the actual labels of the data. It displays the number of true positives (TP), true negatives, false positives (FP), and false negatives (FN) of the model's predictions. The confusion matrix was generated for the test dataset. The test dataset comprised 1,661 safe images and 1,661 unsafe images. This trained model was evaluated based on accuracy, precision, recall, and F1 score metrics. It is observed that the trained model achieved classification performance with an accuracy of 88.7 %, precision of 91.08 %, recall of 85.81 %, and an F1 score of 88.36 %. Due to limitations in the performance approximation method, the accuracy converged with errors.

A total of 1,508 (377×4) image frames were recorded in real-time under practical operating conditions to verify the IQSM, encompassing the four image types (clear, dirty, raindrops on lens, and foggy).

Fig. 16 (left) shows a frame of a grass field scene with a human. In the clear scene, the IQSM predicted to be “safe” with a confidence level of 99.4 %. This indicates that the operating condition meets the minimum safety requirement of the intended function. Notably, YOLO v8, the intended function, successfully detected the human within this scene. However, in dirty, raindrops on lens and foggy scenes, the IQSM outputs a confidence level of 0 % for all cases, resulting in a prediction label “unsafe.” This indicates that the triggering conditions are too poor to perform the intended function safely. Fig. 16 (right) shows saliency maps of each scene. The mean of the saliency map in the clear scene was 0.54 (Fig. 18). Areas containing relatively high-quality features essential for safe intended operation, such as those containing a human, are highlighted in red. Areas containing low-quality features, such as sky

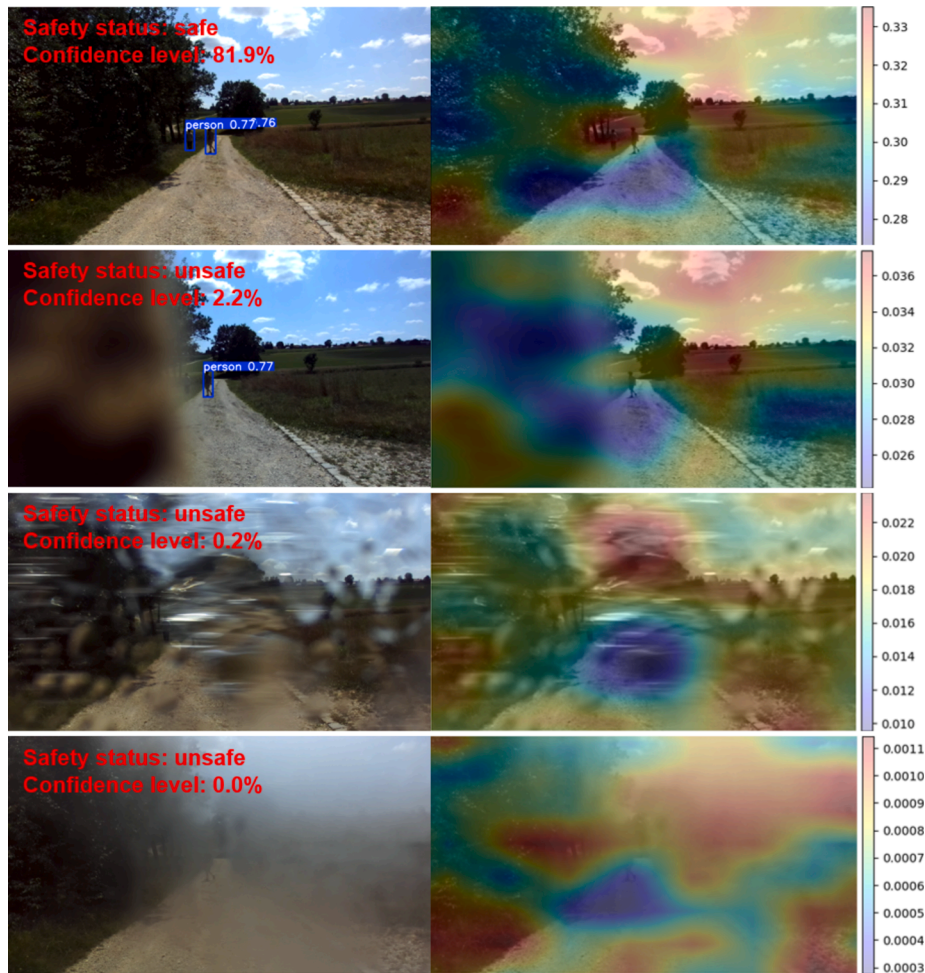


Fig. 17. Scene 2: an off-road scene with clear, dirty, raindrops, foggy conditions and corresponding saliency maps.

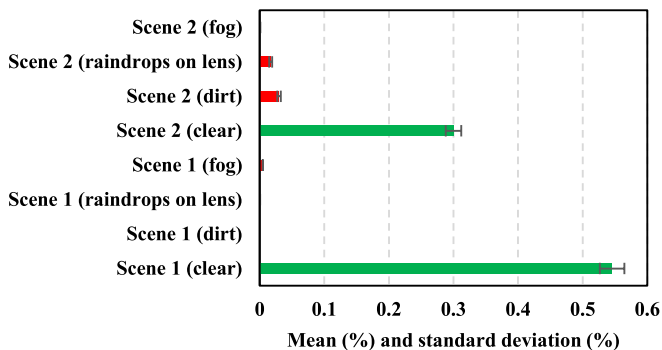


Fig. 18. Summary of saliency maps of scene 1 and scene 2. Red and green bars mean “safe” and “unsafe” status, respectively.

and grass, are shown in blue. However, the mean of saliency maps in dirty, raindrops on lens, and foggy scenes were close to zero, indicating that the overall image quality within these conditions was insufficient for the intended function (Fig. 18).

Fig. 17 (left) shows a frame of an off-road scene with two humans. The IQSM outputs a confidence level of 81.9 %, leading to the prediction label “safe” in this clear scene. The operating condition is sufficient to perform the intended function safely. In fact, two humans were correctly detected by the intended function. However, in cases of dirty, raindrops on lens and foggy scenes, the IQSM outputs confidence levels of 2.2 %, 0.2 %, and 0 %, respectively, resulting in an “unsafe” prediction for all

three. Even though half of the dirty scene was clear enough to detect a human, the overall image quality was too poor to perform the intended function. Fig. 17 (right) shows saliency maps of each scene in the off-road scene. The mean of the saliency map in the clear scene was 0.30 (Fig. 18). Areas containing relatively high-quality features, such as those containing humans, are highlighted in red. Areas containing relatively low-quality features, such as shaded leaves and roads, are shown in blue. The mean of saliency maps in dirty, raindrops on lens, and foggy scenes was 3 %, 1.7 %, and 0 %, respectively, indicating that insufficient image quality for the intended function (Fig. 18).

Fig. 18 shows the summary of saliency maps. In scenes classified as “safe”, the mean of each saliency map was relatively higher than in unsafe scenes. By incorporating the saliency map, the IQSM not only becomes human-interpretable in its decision-making process, but it can also effectively pinpoint and localise safe and unsafe quality regions directly within the image itself. This visual interpretation capability empowers people to understand the reasoning of the model and builds trust in its outputs.

Table 5 presents a summary of the evaluation results for all verification test scenes. The IQSM predicted 97 % of clear scenes as “safe” to perform the intended function. In fact, the intended function made a total of eight errors in all clear scenes: four false detections and four detection failures. This equates to a detection error rate of 2.1 %, meaning that most scenes had enough image quality to safely perform the intended function. Most scenes, including those with dirty raindrops on lens and fog, were classified as “unsafe” by the IQSM. The detection error rates for the conditions were substantial, ranging from 20.9 % to 30.5 %, indicating that many low-quality scenes proved inadequate for

Table 5

Summary of evaluation results of all verification test scenes. Detection error rate, $P(\text{error})$, is the probability of an error occurring. It is calculated as the ratio of false detections and detection failures to the total number of scenes. The malfunction detection rate, $P(\text{unsafe}|\text{error})$, is the probability that an image is unsafe given that an error has occurred. It is calculated using Bayes' theorem as $P(\text{error}|\text{unsafe}) * P(\text{unsafe}) / P(\text{error})$. $P(\text{error}|\text{unsafe})$ is the probability of an error given that an image is unsafe and represents the proportion of errors in unsafe images compared to the total number of unsafe images. $P(\text{unsafe})$ is the probability that an image is unsafe. $P(\text{safe})$ is the probability that an image is safe. Accuracy is the proportion of correct results (both true positives and true negatives) out of the total number of scenes.

Scenarios	Results of prediction of IQSM			Results of intended function			Evaluation of IQSM	
	Safety status	Number of images	Probability [P(safe), P(unsafe)]	Number of images with:		Detection error rate [P(error)]	Malfunction detection rate [P(unsafe error)]	Accuracy $\frac{TP + TN}{TP + TN + FP + FN}$
				False detection	Detection failure			
Clear scenes	Safe	368	0.97	3	4	2.1 %	—	96.0 %
	Unsafe	9	0.02	1	0			
Dirty scenes	Safe	9	0.02	5	1	30.5 %	(too few errors)	
	Unsafe	368	0.97	26	83		94.7 %	97.6 %
Raindrops on lens scenes	Safe	14	0.03	2	1	26.5 %	97 %	99.2 %
	Unsafe	363	0.96	9	88			
Foggy scenes	Safe	45	0.12	0	2	20.9 %	97.4 %	99.4 %
	Unsafe	332	0.88	3	74			

accurate object detection. The IQSM demonstrated significantly higher malfunction detection rates of 94.7 %, 97 %, and 97.4 %, effectively classifying most “unsafe” scenes when an error occurred. The IQSM thus provides a crucial fail-safe mechanism. Moreover, the IQSM demonstrated exceptional accuracy in classifying entire test scenes, achieving rates between 96 % and 99.4 %.

The identification and mitigation of IQSM limitations are part of the SOTIF process for reducing areas 2 and 3 (Fig. 2) (ISO, 2022). These activities are handled by verification and validation.

4. Conclusions

This article describes the concept and method for developing an Image Quality Safety Model (IQSM) for the safety of the intended functionality. Data processing, labelling, training, and test methods are detailed for developing the IQSM. Finally, the trained IQSM is evaluated and validated based on foreseeable hazardous scenarios, such as dirt, raindrops on lens, and fog.

The following conclusions can be drawn:

- The IQSM can estimate the confidence level in whether a camera can perform an object detection function safely. If the confidence level falls below the minimum performance threshold, the IQSM can notice unsafe situations. Therefore, the IQSM can prevent malfunctions, such as failure to detect an obstacle, in perception systems and enables the safety of the intended functionality.
- The IQSM makes decisions based on the quality of the image rather than the presence or absence of objects in an environment. This makes it applicable even in environments where objects are not present.
- The use of various image degradation techniques proved instrumental in constructing training datasets for a model capable of identifying functional insufficiencies in an intended function.
- The trained IQSM achieved a validation accuracy of about 90 %, indicating that it learned distinctive image quality features in different conditions. A confusion matrix generated for a test dataset showed the classification performance of the trained IQSM, with an accuracy of 88.7 %, precision of 91.08 %, recall of 85.81 %, and an F1 score of 88.36 %.
- The IQSM demonstrated exceptional performance in evaluating the suitability of validation test scenes for object detection. In clear scenes, the IQSM accurately classified 97 % as “safe,” with a low detection error rate of 2.1 %. Conversely, the IQSM effectively identified “unsafe” conditions in scenes with low image quality, such as those with dirt, raindrops on lens, or fog. While the detection error rates for these conditions were higher, ranging from 20.9 % to 30.5

%, the malfunction detection rates of the IQSM were significantly elevated, reaching 94.7 % to 97.4 %. Overall, the accuracy of the IQSM for entire test scenes was consistently high, ranging from 98.1 % to 99.4 %, demonstrating its effectiveness in assessing the overall quality of the environment.

- The IQSM is essential for identifying discrepancies between input data distribution during the runtime of an intended function and the distribution of data utilised in the training phase. It helps to systematically mitigate the risks of known hazardous behaviour related to SOTIF. This IQSM enables the implementation of fail-safe architectures in diverse applications, including highly automated agricultural machinery.
- The current performance approximation method during the labelling phase has limitations in accurately representing the relationship between image quality and the performance of the intended function.
- The IQSM provides a representative value indicating the quality of the entire image. For improved predictive accuracy or specific applications, it is more effective to focus on an area of interest or to take a smaller, random sample from the image, rather than processing the entire image at once.
- The basic development process of the IQSM can be generalized to a data quality safety model for the safety of the intended functions of various perception sensors beyond the safety of camera-based object detection systems. For example, a model could support the safety of LiDAR-based object detection.
- Future work will focus on implementation of a fault management system capable of addressing hazardous behaviours in perception systems from both functional safety and SOTIF perspectives.

CRedit authorship contribution statement

Changjoo Lee: Writing – original draft, Software, Methodology, Conceptualization. **Simon Schätzle:** Writing – review & editing, Supervision. **Stefan Andreas Lang:** Writing – review & editing, Supervision. **Timo Oksanen:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [Changjoo Lee reports financial support was provided by Federal Ministry of Education and Research Bonn Office. Changjoo Lee has patent pending to Assignee. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.].

Acknowledgement

This research was supported by the KI4BoardNet project (16ME0765), which is funded by the Federal Ministry of Education and Research (BMBF), Federal Republic of Germany.

Data availability

The authors do not have permission to share data.

References

- Aqqa, M., Mantini, P., Shah, S.K., 2019. Understanding How Video Quality Affects Object Detection Algorithms, VISIGRAPP, pp. 96–104, doi: 10.5220/0007401600960104.
- Berrar, D., 2019. Cross-validation, doi: 10.1016/B978-0-12-809633-8.20349-X.
- Board, C.A., 2023. IEEE Standard for Camera Phone Image Quality. IEEE Standard, <https://standards.ieee.org/ieee/1858/6931/>.
- Borel-Donohue, C., Young, S.S., 2019. Image quality and super resolution effects on object recognition using deep neural networks, Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications. SPIE, pp. 596–604, doi: 10.1117/12.2518524.
- Christiansen, P., 2017. TractorEYE: vision-based real-time detection for autonomous vehicles in agriculture. doi: 10.7146/aul.277.192.
- Chen, Z., Lin, W., Wang, S., Xu, L., Li, L., 2017. Image quality assessment guided deep neural networks training. doi: 10.48550/arXiv.1708.03880.
- Dodge, S., Karam, L., 2016. Understanding how image quality affects deep neural networks, 2016 eighth international conference on quality of multimedia experience (QoMEX). IEEE, pp. 1–6, doi: 10.48550/arXiv.1604.04004.
- Doran, M., Sterritt, R., Wilkie, G., 2020. Autonomic architecture for fault handling in mobile robots. Innov. Syst. Softw. Eng. 16, 263–288. <https://doi.org/10.1007/s11334-020-00361-8>.
- Edan, Y., Han, S., Kondo, N., 2009. Automation in agriculture. Springer Handbook of Automation 1095–1128. https://doi.org/10.1007/978-3-540-78831-7_63.
- Goelles, T., Schlager, B., Muckenhuber, S., 2020. Fault detection, isolation, identification and recovery (fdiir) methods for automotive perception sensors including a detailed literature survey for lidar. Sensors 20, 3662. <https://doi.org/10.3390/s20133662>.
- Goutte, C., Gaussier, E., 2005. A probabilistic interpretation of precision, recall and F-score, with implication for evaluation, European conference on information retrieval. Springer, pp. 345–359, doi: 10.1007/978-3-540-31865-1_25.
- Group, I.P.W., 2018. IEEE P2020 Automotive Imaging: White Paper. IEEE P2020 Automotive Imaging. IEEE-SA (IEEE, Piscataway, NJ, 2018), https://www.image-engineering.de/content/library/white_paper/P2020_white_paper.pdf.
- Gulli, A., Kapoor, A., Pal, S., 2019. Deep learning with TensorFlow 2 and Keras: regression, ConvNets, GANs, RNNs, NLP, and more with TensorFlow 2 and the Keras API. Packt Publishing Ltd.
- IEC, 2018. Video surveillance systems for use in security applications, Part 5: Data specifications and image quality performance for camera devices. IEC 62676-5:2018.
- ISO, 2017. Earth-moving machinery — Object detection systems and visibility aids — Performance requirements and tests. ISO 16001:2017, Switzerland.
- ISO, 2018a. Agricultural machinery and tractors — Safety of highly automated agricultural machines — Principles for design. ISO 18497:2018, Switzerland.
- ISO, 2018b. Tractors and machinery for agriculture and forestry — Safety-related parts of control systems. ISO 25119-1:2018, Switzerland.
- ISO, 2019. Earth-moving machinery and mining — Autonomous and semi-autonomous machine system safety. ISO 17757:2019, Switzerland.
- ISO, 2022. Road vehicles — Safety of the intended functionality. ISO 21448:2022, Switzerland.
- ISO, 2024. Artificial intelligence — Functional safety and AI systems. ISO/IEC TR 5469:2024, Switzerland.
- Klås, M., Vollmer, A.M., 2018. Uncertainty in machine learning applications: A practice-driven classification of uncertainty, Computer Safety, Reliability, and Security: SAFECOMP 2018 Workshops, ASSURE, DECSOs, SASSUR, STRIVE, and WAISE, Västerås, Sweden, September 18, 2018, Proceedings 37. Springer, pp. 431–438, doi: 10.1007/978-3-319-99229-7_36.
- Kong, L., Ikusan, A., Dai, R., Zhu, J., 2019. Blind image quality prediction for object detection. In: 2019 IEEE Conference on Multimedia Information Processing and Retrieval (MIPR). IEEE, pp. 216–221. <https://doi.org/10.1109/mipr.2019.00046>.
- Lee, C., Schätzle, S., Lang, S.A., 2022. Evaluation of perception sensors for safety of highly automated machine operations. VDI Verlag eBooks 163–170. <https://doi.org/10.51202/9783181023952-163>.
- Lee, C., Schätzle, S., Lang, S.A., Oksanen, T., 2023. Design considerations of a perception system in functional safety operated and highly automated mobile machines. Smart Agric. Technol. 6, 100346. <https://doi.org/10.2139/ssrn.4425819>.
- Li, Y., Liu, L., 2019. Image quality classification algorithm based on InceptionV3 and SVM, MATEC Web of Conferences. EDP Sciences 02036. <https://doi.org/10.1051/mateconf/201927702036>.
- Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., Zitnick, C.L., 2014. Microsoft coco: Common objects in context, Computer Vision—ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, 2014, Proceedings, Part V 13. Springer, pp. 740–755, <https://arxiv.org/abs/1405.0312>.
- Linenko, A.V., Gabitov, I.I., Siraev, S.F., Aznagulov, A.I., Lukyanov, V.V., Kamalov, T.I., 2019. Farm tractor mechatronic steering module. J. Appl. Eng. Sci. 17. <https://doi.org/10.5937/jaes17-22052>.
- Marti, E., De Miguel, M.A., Garcia, F., Perez, J., 2019. A review of sensor technologies for perception in automated driving. IEEE Intell. Transp. Syst. Mag. 11, 94–108. <https://doi.org/10.1109/mits.2019.2907630>.
- Petsiuk, V., 2018. RISE: Randomized Input Sampling for Explanation of black-box models. doi: 10.48550/arXiv.1806.07421.
- Porav, H., Bruls, T., Newman, P., 2019. I can see clearly now: Image restoration via de-raining. In: 2019 International Conference on Robotics and Automation (ICRA). IEEE, pp. 7087–7093. <https://doi.org/10.1109/icra.2019.8793486>.
- Portugal, D., Cabrita, G., Gouveia, B.D., Santos, D.C., Prado, J.A., 2015. An autonomous all terrain robotic system for field demining missions. Rob. Auton. Syst. 70, 126–144. <https://doi.org/10.1016/j.robot.2015.02.013>.
- Reis, D., Kupec, J., Hong, J., Daoudi, A., 2023. Real-time flying object detection with YOLOv8. arXiv preprint arXiv:2305.09972, doi: 10.48550/arXiv.2305.09972.
- Sebe, N., Lew, M.S., 2013. Robust computer vision: Theory and applications. Springer Science & Business Media.
- Soboleva, V., Shipitko, O., 2021. Raindrops on windshield: Dataset and lightweight gradient-based detection algorithm, 2021 IEEE Symposium Series on Computational Intelligence (SSCI). IEEE, pp. 1–7, <https://ieeexplore.ieee.org/document/9659915>.
- Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., Wojna, Z., 2016. Rethinking the inception architecture for computer vision. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 2818–2826. <https://doi.org/10.48550/arXiv.1512.00567>.
- Tiusanen, R., Malm, T., Ronkainen, A., 2020. An overview of current safety requirements for autonomous machines—review of standards. Open Eng. 10, 665–673. <https://doi.org/10.1515/eng-2020-0074>.
- Zhang, Q., 2016. Precision agriculture technology for crop farming. Taylor & Francis.